

1B. Dot Product

1B-1 Find the angle between the vectors:

- (a) $\mathbf{i} - \mathbf{k}$ and $4\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$
- (b) $\mathbf{i} + \mathbf{j} + 2\mathbf{k}$ and $2\mathbf{i} - \mathbf{j} + \mathbf{k}$

1B-2 Tell for what values of c the vectors $c\mathbf{i} + \mathbf{j} - \mathbf{k}$ and $\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ will be:

- (a) Orthogonal
- (b) Form an acute angle

1B-3 Using vectors, find the angle between the longest diagonal PQ of a cube and:

- (a) A diagonal PR of one of its faces
- (b) An edge PS of the cube

(Choose a size and position for the cube that makes calculation easiest.)

1B-4 Three points in space are

$$P(a, 1, -1), \quad Q(0, 1, 1), \quad R(a, -1, 3).$$

For what value(s) of a will $\angle PQR$ be:

- (a) A right angle
- (b) An acute angle

1B-5 Find the component of the force

$$\mathbf{F} = 2\mathbf{i} - 2\mathbf{j} + \mathbf{k}$$

in:

- (a) The direction $\frac{\mathbf{i} + \mathbf{j} - \mathbf{k}}{\sqrt{3}}$
- (b) The direction of the vector $3\mathbf{i} + 2\mathbf{j} - 6\mathbf{k}$

1. Vectors and Matrices

1B-6 Let O be the origin, c a given number, and $\mathbf{u}$ a given direction (i.e. a unit vector). Describe geometrically the locus of all points P in space that satisfy the vector equation

$$\overrightarrow{OP} \cdot \mathbf{u} = c |\overrightarrow{OP}|.$$

In particular, tell for what value(s) of c the locus will be:

- (a) A plane
- (b) A ray (i.e. a half-line)
- (c) Empty

1B-7

- (a) Verify that

$$\mathbf{i}' = \frac{\mathbf{i} + \mathbf{j}}{\sqrt{2}} \quad \text{and} \quad \mathbf{j}' = \frac{-\mathbf{i} + \mathbf{j}}{\sqrt{2}}$$

are perpendicular unit vectors that form a right-handed coordinate system.

- (b) Express the vector

$$\mathbf{A} = 2\mathbf{i} - 3\mathbf{j}$$

in the $(\mathbf{i}', \mathbf{j}')$ system using the dot product.

- (c) Do part (b) in a different way by solving for $\mathbf{i}$ and $\mathbf{j}$ in terms of $\mathbf{i}'$ and $\mathbf{j}'$, and then substituting into the expression for $\mathbf{A}$.

1B-8 The vectors

$$\mathbf{i}' = \frac{\mathbf{i} + \mathbf{j} + \mathbf{k}}{\sqrt{3}}, \quad \mathbf{j}' = \frac{\mathbf{i} - \mathbf{j}}{\sqrt{2}}, \quad \mathbf{k}' = \frac{\mathbf{i} + \mathbf{j} - 2\mathbf{k}}{\sqrt{6}}$$

are three mutually perpendicular unit vectors that form a right-handed coordinate system.

1B-9 Let $\mathbf{A}$ and $\mathbf{B}$ be two plane vectors, neither one of which is a multiple of the other. Express $\mathbf{B}$ as the sum of two vectors, one a multiple of $\mathbf{A}$ and the other perpendicular to $\mathbf{A}$. Give the answer in terms of $\mathbf{A}$ and $\mathbf{B}$.

1B-10 Prove using vector methods (without components) that the diagonals of a parallelogram have equal lengths if and only if it is a rectangle.

1B-11 Prove using vector methods (without components) that the diagonals of a parallelogram are:

- (a) Equal in length if and only if it is a rectangle;
- (b) Perpendicular if and only if it is a rhombus (i.e. its four sides are equal).

1B-12 Prove using vector methods (without components) that an angle inscribed in a semicircle is a right angle.

Vector and Trigonometric Identities

1. Trigonometric Formula

Prove that:

$$\cos(\theta_1 - \theta_2) = \cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2$$

2. Law of Cosines

Prove that:

$$c^2 = a^2 + b^2 - 2ab \cos \theta$$

3. Cauchy–Schwarz Inequality

Show that:

$$|A \cdot B| \leq |A||B|$$

If $A = (a_1, a_2)$ and $B = (b_1, b_2)$, then

$$|a_1 b_1 + a_2 b_2| \leq \sqrt{a_1^2 + a_2^2} \sqrt{b_1^2 + b_2^2}$$

Determinants

1C-1

Calculate the value of the determinant:

a)

$$\begin{vmatrix} 1 & 4 \\ 2 & -1 \end{vmatrix}$$

b)

$$\begin{vmatrix} 3 & -4 \\ -1 & 2 \end{vmatrix}$$

1C-2

Calculate

$$\begin{vmatrix} -1 & 0 & 4 \\ 1 & 2 & 2 \\ 3 & -2 & -1 \end{vmatrix}$$

using the Laplace expansion:

- (a) Along the first row
- (b) Along the first column

1C-3

Find the area of the plane triangle whose vertices are:

- (a) $(0, 0)$, $(4, 2)$, $(1, -1)$
- (b) $(1, 2)$, $(4, 1)$, $(2, 3)$

1C-4

Show that

$$\begin{vmatrix} 1 & 1 & 1 \\ x_1 & x_2 & x_3 \\ x_1^2 & x_2^2 & x_3^2 \end{vmatrix} = (x_1 - x_2)(x_2 - x_3)(x_3 - x_1)$$

This type of determinant is called a **Vandermonde determinant**.

1C-5

- (a) Show that the value of a 2×2 determinant is unchanged if you add to the second row a scalar multiple of the first row.
- (b) Same question with “row” replaced by “column”.

1C-6

Use Laplace expansion and Exercise 5(a) to show that the value of a 3×3 determinant is unchanged if you add to the second row a scalar multiple of the third row.

1C-7

Let (x_1, y_1) and (x_2, y_2) both range over all unit vectors. Find the maximum value of the function

$$f(x_1, x_2, y_1, y_2) = \begin{vmatrix} x_1 & y_1 \\ x_2 & y_2 \end{vmatrix}.$$

1C-8

The base of a parallelepiped is a parallelogram whose edges are the vectors $\mathbf{b}$ and $\mathbf{c}$, while its third edge is the vector $\mathbf{a}$.

(All three vectors have their tails at the same vertex; one calls them coterminal.)

Show that the volume of the parallelepiped abc is

$$V = |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|.$$